Level 1 Practice Programs

Java program solutions without comments

Each question is shown with a clean, compile-ready solution.

Question 1: Simple Interest

```
import java.util.Scanner;

public class SimpleInterestProgram {
    public static double calculateSimpleInterest(double principal, double rate, double time) {
        return (principal * rate * time) / 100.0;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Principal: ");
        double principal = sc.nextDouble();
        System.out.print("Enter Rate: ");
        double rate = sc.nextDouble();
        System.out.print("Enter Time: ");
        double time = sc.nextDouble();

        double si = calculateSimpleInterest(principal, rate, time);
        System.out.println("The Simple Interest is " + si
                + " for Principal " + principal
                + ", Rate of Interest " + rate
                + " and Time " + time);
        sc.close();
    }
}
```

Question 2: Maximum Handshakes

```
import java.util.Scanner;

public class HandshakeProgram {
    public static long calculateHandshakes(long numberOfStudents) {
        return numberOfStudents * (numberOfStudents - 1) / 2;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of students: ");
        long numberOfStudents = sc.nextLong();

        long handshakes = calculateHandshakes(numberOfStudents);
        System.out.println("Maximum possible handshakes: " + handshakes);
        sc.close();
    }
}
```

Question 3: Athlete Running in Triangular Park

```
import java.util.Scanner;

public class TriangularParkRun {
    public static double calculateRounds(double side1, double side2, double side3) {
        double perimeter = side1 + side2 + side3;
        return Math.ceil(5000.0 / perimeter);
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter side 1 in meters: ");
        double side1 = sc.nextDouble();
        System.out.print("Enter side 2 in meters: ");
        double side2 = sc.nextDouble();
        System.out.print("Enter side 3 in meters: ");
        double side3 = sc.nextDouble();

        double rounds = calculateRounds(side1, side2, side3);
        System.out.println("Number of rounds required to complete 5 km run: " + rounds);
        sc.close();
    }
}
```

Question 4: Positive, Negative, or Zero

```
import java.util.Scanner;

public class NumberSignProgram {
    public static int checkNumber(int number) {
        if (number > 0) {
            return 1;
        } else if (number < 0) {
            return -1;
        }
        return 0;
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter a number: ");
        int number = sc.nextInt();

        int result = checkNumber(number);
        if (result == 1) {
            System.out.println("Positive number");
        } else if (result == -1) {
            System.out.println("Negative number");
        } else {
            System.out.println("Zero");
        }
        sc.close();
    }
}
```